

The 24th - Game Design Document

v0.2

Working Title: The 24th

Genre: First-Person Arena Survival / Survival Combat / PvE-PvNPC

Engine: Unreal Engine 5.7

Development Style: Blueprint-first, with Codex + NoGlyph Blueprint Stack assisting Unreal Editor automation, Blueprint generation, and project implementation

Platform Target: PC / Steam Early Access first

Camera Target: First-person for v1 prototype, optional third-person later

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Document Status: Early concept / planning draft

1. Project Overview

The 24th is a first-person arena survival game where 24 Survivors are raised into a sealed biome Dome and forced into a deadly survival Trial. The player is one of the 24 Survivors. The remaining 23 are NPC competitors with different behaviours, survival patterns, combat styles, fears, and strategies. The match begins with the player standing inside a sealed underground elevator pod. A countdown begins. The platform rises into the arena. The Dome opens into view. Other Survivors rise on distant platforms around the centre. At the heart of the arena is The Core, a high-risk supply zone containing weapons, food, tools, and survival items. The player may rush The Core for an early advantage or flee into the wilderness and survive through crafting, hunting, stealth, scavenging, and careful decision-making. The Dome is controlled by an unseen system known as The Operators, who manipulate the arena through supply drops, weather shifts, zone warnings, wildlife releases, drones, and environmental hazards.

The goal is simple:

Survive the Dome. Outlast 23 others. Become the Ascendant.

2. Core Identity

2.1 Core Sentence

The 24th is a first-person arena survival game where 24 Survivors rise into a sealed Dome and must craft, hunt, hide, fight, and adapt until only one remains.

2.2 Core Hook

The player is dropped into a contained, hostile arena with 23 NPC Survivors. Every match begins the same way - rising into the Dome - but unfolds differently depending on loot choices, NPC behaviour, arena events, weather, wildlife, and player strategy.

2.3 Player Fantasy

The player should feel like a desperate survivor inside a controlled death arena. They are not a superhero, soldier, or chosen warrior. They are under-equipped, watched, hunted, and forced to adapt.

The fantasy is:

Rising into the arena with no certainty of survival. Choosing whether to fight, flee, hide, hunt, craft, or scavenge. Tracking other Survivors through footprints, sounds, smoke, dropped items, and movement.

Surviving hunger, thirst, injury, stamina loss, wildlife, traps, and environmental pressure. Slowly becoming more capable through better gear, smarter choices, and experience. Reaching the final moments with only a few Survivors left. Escaping as the last one alive.

3. Legal / IP-Safe Creative Direction

The project is inspired by the broad concept of a deadly contained survival arena, but it must remain original in naming, lore, character identity, presentation, visual design, and terminology. The game must avoid directly copying characters, faction systems, social structures, symbols, slogans, scene layouts, and iconic franchise terminology from existing arena-survival fiction.

3.1 Terms to Avoid

The following terms should not be used in the project's public-facing lore, UI, marketing, quests, naming systems, or game dialogue:

Tributes

Districts

Capitol

Panem

Reaping

Mockingjay

Peacekeepers

Cornucopia

Gamemakers

Career tributes

Quarter Quell

Victor's Village President Snow Any direct character names from existing source material

3.2 Approved Original Terms

Concept

Original Term for The 24th

Contestants / players

Survivors

Winner

Ascendant

Arena

The Dome

Match

The Trial

Centre loot zone

The Core

Arena border

Boundary Field

Arena controllers
The Operators
Match announcements
Broadcasts
Supply drops
Drops
Mutated/hostile arena creatures
Bioforms
Entry process
Intake
Selection process
Selection
Final escape point
The Gate
Final extraction sequence
Ascension

3.3 Tone Rules

The game should not feel like a political rebellion story. It should be focused on arena survival first.

Avoid:

A direct rebellion plotline. A president villain figure. A district-based society structure. A bow-wielding heroine identity. Famous salute/symbolism parallels. Romance-driven survival arcs. Exact scene recreations from films or books.

Use instead:

A mysterious controlled arena.

Minimalist dystopian background. Environmental storytelling. Survivor-focused character design. AI/control-system atmosphere. Arena systems that feel cold, mechanical, and unfair.

4. Creative Pillars

The whole game should be designed around these pillars.

4.1 Survival First, Combat Second

Combat should be dangerous. The player should not feel encouraged to fight every Survivor immediately. Hunting, hiding, crafting, ambushing, fleeing, and reading the environment should be just as valid as direct combat.

4.2 The Arena Is the Main Enemy

The Dome should feel alive and hostile. The player is not only fighting other Survivors; they are fighting hunger, thirst, stamina, weather, terrain, traps, wildlife, visibility, and the arena's manipulation systems.

4.3 Every Match Tells a Story

Each match should produce emergent moments:

A Survivor fleeing wounded through the forest. Smoke rising from a hidden campfire. A trap placed near a supply cache. A storm forcing everyone into the same region. A final duel near the Boundary Field.

4.4 Clear Rules, Unpredictable Outcomes

The player should understand the rules, but not know exactly what will happen each match. NPC behaviour, loot distribution, wildlife spawns, arena events, and environmental conditions should create replayability.

4.5 No Base Building

The game should include crafting and temporary survival actions, but not permanent base building. This keeps the pacing focused on movement, tension, and the match structure.

4.6 Build Modular Systems

Every major gameplay feature should be developed as a modular Blueprint system so Codex, connected through NoGlyph Blueprint Stack, can inspect the Unreal project, generate or edit Blueprints, create variables, wire node graphs, assist with UI logic, and help maintain clean implementation across the project.

5. Game Modes

5.1 Primary Mode: Single-Player Trial

The first version should be single-player only. 1 player Survivor. 23 NPC Survivors. One Dome map. One full match from elevator intro to victory/death. This is the most achievable version and should be the first vertical slice target.

5.2 Future Mode: Custom Trial

Custom Trial allows the player to adjust match settings.

Possible settings:

Number of Survivors. Loot abundance. Wildlife danger. Hunger/thirst drain speed. Event frequency. Dome shrink speed. Difficulty level. Starting gear.

5.3 Future Mode: Multiplayer Trial

Multiplayer should be considered later, not for the first prototype.

Possible formats:

1 player + 23 NPCs. 4 players + 20 NPCs. 24-player PvP long-term. Co-op survival variant with shared extraction objective. Multiplayer will significantly increase complexity, so it should only be explored after the core single-player systems are fun.

6. Narrative Direction

6.1 Story Scope

The story should be minimal and arena-focused. The game does not need a large political background at first.

The player only needs to understand:

They have been selected. They are one of 24 Survivors. The Dome is sealed. The Trial is watched or controlled. Only one Survivor becomes the Ascendant.

6.2 World Premise

In a damaged future, controlled survival Trials are used to select those deemed worthy of entry into protected society. Each Trial takes place inside a sealed ecological Dome. The Dome contains natural biomes, artificial hazards, wildlife, resources, hidden supply systems, and controlled environmental events. The Survivors are raised into the Dome from below. Once the Trial begins, the Boundary Field activates and escape becomes impossible. The Operators observe and manipulate the Trial until only one Survivor remains.

6.3 The Operators

The Operators are the unseen controllers of the Dome. They do not need visible characters at first. They can

exist through:

Broadcast voice lines. Dome warning sirens. Supply drop announcements. Event warnings. UI notifications. Environmental changes. Drone activity. Their role is to create tension and pressure.

6.4 The Ascendant

The winner of the Trial is called the Ascendant. This term can have both gameplay and lore meaning.

Gameplay meaning:

The last surviving player/NPC.

Lore meaning:

The one who has proven survival value. The one extracted from the Dome. The one granted passage beyond the Boundary.

7. Core Gameplay Loop

The full match loop:

Prepare - Player enters the elevator pod and sees basic instructions/loading narrative. Ascend - Elevator rises into the Dome. Decide - Rush The Core or flee into the wilderness. Scavenge - Collect weapons, food, tools, medicine, materials, and water. Survive - Manage health, hunger, thirst, stamina, injuries, weather, and wildlife. Craft - Create tools, weapons, traps, bandages, and survival items. Hunt / Hide / Fight - Avoid, track, ambush, or confront other Survivors. Adapt - Respond to arena events, drops, zone shifts, and environmental hazards. Outlast - Survive until only a few Survivors remain. Final Pressure - Dome events force the remaining Survivors closer together. Ascend - Defeat/outlast the final Survivor and reach extraction.

7.1 Minute-to-Minute Gameplay

Move through terrain. Listen for movement, wildlife, or combat. Gather small resources. Check vitals. Loot containers or corpses. Hide in foliage or terrain. Track Survivor signs. Craft quick survival tools. Avoid exposure. Decide whether to engage or retreat.

7.2 Short-Term Goals

Find water. Find food. Obtain a weapon. Avoid early death. Craft bandages. Reach a safer zone. Avoid stronger Survivors. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11.

7.3 Mid-Term Goals

Upgrade weapon quality. Build basic trap inventory. Track NPC movement. Secure healing items. Hunt animals. Learn zone routes. Avoid dangerous arena events.

7.4 Long-Term Goals

Survive until the final phase. Eliminate or outlast major threats. Maintain enough supplies for the final confrontation. Reach The Gate after becoming the last Survivor.

8. Match Structure

8.1 Pre-Match Loading

The player loads into a dark preparation space or black-screen sequence with audio.

Possible audio elements:

Mechanical hum. Elevator locking mechanisms. Distant announcement. Heartbeat/breathing. Countdown.

8.2 Elevator Intro

The player starts inside a small vertical pod.

Sequence:

Player has limited movement/camera control. Lights flicker on. UI shows "Survivor 24" or player designation. Countdown begins. Elevator rises. Bright light floods the pod. The Dome is revealed. Other platforms rise in the distance. The Core becomes visible. 1. 2. 3. 4. 5. 6. 7. 8. 9.

Match begins when the alarm sounds.

8.3 Opening Phase: The First Rush

The opening creates immediate tension.

Player choices:

Rush The Core for better loot. Grab one nearby item and flee. Ignore centre loot and run into a biome. Hide and observe. Follow another Survivor.

NPC choices:

Some rush The Core. Some flee immediately. Some attack nearest targets. Some grab loot and retreat. Some freeze or panic briefly.

8.4 Early Game

Focus:

Basic survival. Avoiding stronger enemies. Finding food/water. Crafting first tools. Learning the zone. Duration target: first 15-25% of match.

8.5 Mid Game

Focus:

NPCs spread out. Wildlife becomes more relevant. Survivors begin crafting and tracking. Supply drops begin. Arena events escalate. Duration target: middle 50% of match. 10.

8.6 Late Game

Focus:

Fewer Survivors remain. Resources become more contested. Dome events push Survivors toward conflict. The player must be prepared for combat. Duration target: final 20-30% of match.

8.7 Final Phase

Focus:

Remaining Survivors are revealed or pressured inward. Weather/event intensity increases. The Gate powers up only after one Survivor remains. Player must defeat/outlast the final threat.

8.8 Victory Sequence

When the player becomes the last Survivor:

Arena siren stops. Broadcast announces final Survivor status. The Gate activates. Player follows marker to extraction. Boundary Field opens in one location. Player exits the Dome or enters extraction lift. Victory screen appears.

8.9 Death Sequence

When the player dies:

Camera falls or fades. Remaining Survivor count appears. Killer name/archetype may be shown. Match summary appears. Option: Restart Trial / Return to Menu. No respawn in standard mode. 1. 2. 3. 4. 5. 6. 7.

9. Arena Design

9.1 Main Arena: The Dome

The Dome is a large sealed arena containing several distinct biomes and landmarks. It should feel natural at first glance, but with artificial boundaries, hidden systems, and controlled event triggers.

The Dome should be designed around circular progression:

Centre: high risk / high loot. Middle zones: survival, hunting, crafting, stealth. Outer zones: dangerous border pressure and hidden resources. Event pressure: forces Survivors inward or toward specific areas.

9.2 Arena Scale

Early prototype should use a smaller test arena before building a full-scale map.

Suggested test sizes:

Prototype Arena: 500m-800m diameter. Vertical Slice Arena: 1km-1.5km diameter. Full Arena: 2km-3km diameter, depending on performance. The first version should prioritise density over size. A smaller arena with meaningful landmarks is better than a massive empty map.

9.3 The Core

The Core replaces the idea of a central loot object with something original.

The Core is a large circular mechanical structure at the centre of the Dome. It may contain:

Supply crates. Weapon racks. Medical containers. Water canisters. Food packs. Crafting materials. Locked containers. Rare items. Alarm systems. The Core should be dangerous because many Survivors are drawn to it.

9.4 Arena Zones

Zone 01 - The Core Ring

Central starting area.

Features:

Elevator platforms. The Core structure. Highest early-game loot. Exposed sightlines. Immediate danger.

Gameplay purpose:

Creates opening decision. Creates early conflict. Establishes match tension. Zone 02 - Pine Belt Dense forest region.

Features:

Trees. Bushes. Deer/rabbit-like wildlife. Wood harvesting. Plant gathering. Hiding opportunities.

Gameplay purpose:

Stealth. Hunting. Basic crafting resources. Zone 03 - Dry Ridge Rocky high-ground region.

Features:

Cliffs. Caves. Stone nodes.

Ambush points. Long sightlines.

Gameplay purpose:

Bow combat. Tracking. Risky traversal. High-ground strategy. Zone 04 - Marsh Hollow Wetland region.

Features:

Shallow water. Reeds. Mud. Slower movement. Herbs. Insects/snakes. Hidden water risks.

Gameplay purpose:

Healing resources. Water access. Stealth through reeds. Movement penalties. Zone 05 - The Ruins Old abandoned structures inside the Dome.

Features:

Broken concrete/metal structures. Rare loot. Traps. Hidden containers. Environmental storytelling.

Gameplay purpose:

Mid-game objective zone. Dangerous exploration. Lore hints.

Zone 06 - Boundary Verge

Outer edge near the Dome wall.

Features:

Visual distortion. Electric hum. Limited resources. High danger. Dome wall effects.

Gameplay purpose:

Prevents escape. Creates pressure. Can hide rare resources but carries risk.

10. Dome and Boundary Field

10.1 Boundary Field Purpose

The Boundary Field prevents Survivors from leaving the arena.

It should communicate danger clearly:

Audio hum increases near border. Screen distortion appears. UI warning appears. Health damage begins if player stays too close. Instant death only if player pushes too far or ignores warnings.

10.2 Boundary Field Mechanics

Variables:

BoundaryWarningDistance

BoundaryDamageDistance

BoundaryKillDistance

BoundaryDamagePerSecond

blsInWarningZone blsTakingBoundaryDamage

Player feedback:

Warning: "Boundary Field unstable."

Damage: electrical distortion, controller/camera shake. Critical: red UI pulse and audio spike.

10.3 Dome Shrink / Pressure System

Optional for later, but valuable. The Dome itself does not need to physically shrink. Instead, danger zones can activate to push Survivors inward.

Possible systems:

Storm zones. Toxic fog. Thermal sweep. Drone scan. Wildlife release. Boundary surge. This avoids feeling like a battle royale clone while still creating pressure.

11. Player Character

11.1 Player Role

The player is Survivor 24 by default. This supports the title The 24th. The game can later allow character customisation, but v0.1 can use one default character.

11.2 Camera

Version 1 target:

First-person camera. Full-body awareness optional later. Hands visible for weapons/tools. Head bob subtle. Camera shake for impacts, exhaustion, and boundary damage.

Future optional:

Third-person toggle. Photo mode.

11.3 Basic Controls

Suggested default controls:

Action

Key

Move

WASD

Look

Mouse

Jump

Space

Sprint

Left Shift

Crouch

Left Ctrl

Interact

E Hold Interact Hold E Inventory Tab Crafting C Map / Mini-map Expand M Hotbar Slots 1-6 Attack Left Mouse Aim / Block Right Mouse Use Item F Drop Item X Toggle UI Info V Pause Esc

12. Survival Stats System

12.1 Core Stats

The player has the following core stats:

Health

Stamina

Hunger

Thirst

Potential later stats:

Body Temperature

Infection

Bleeding

Pain

Noise Level

Visibility Level

12.2 Health

Health represents the player's life.

Health can be reduced by:

Melee attacks. Ranged attacks. Wildlife attacks. Traps. Falling. Starvation. Dehydration. Boundary Field damage. Poison/infection later.

Health can be restored by:

Bandages. Herbal poultices. Medical packs. Cooked food small bonus. Rest/campfire later.

12.3 Stamina

Stamina controls sprinting, heavy attacks, jumping, climbing, and emergency actions.

Stamina drains when:

Sprinting. Jumping. Climbing. Heavy attacking. Dodging later.

Stamina regenerates when:

Walking.

Standing still. Crouching. Resting.

Low stamina effects:

Slower sprint speed. Heavy breathing. Reduced attack speed. Poor aim stability. Screen vignette.

12.4 Hunger

Hunger gradually decreases over time.

Effects at low hunger:

Reduced stamina regeneration. Slower movement over time. Health damage at zero.

Food sources:

Berries. Mushrooms. Cooked meat. Raw meat with risk. Packaged food from loot containers.

12.5 Thirst

Thirst decreases faster than hunger.

Effects at low thirst:

Reduced stamina maximum. Blurred vision. Health damage at zero.

Water sources:

Streams. Rain collection later. Water canisters. Marsh water requiring purification. Water pouch.

12.6 Injury States

Initial injury states:

Bleeding

Broken/strained leg

Poisoned

Exhausted

Bleeding:

Slowly drains health. Leaves blood trail for tracking. Fixed with bandage.

Leg injury:

Reduces movement speed. Can happen from falls/traps. Fixed with splint or medical item.

Poisoned:

Caused by wildlife/plants. Slowly drains health or stamina. Fixed with antidote/herbal mix.

Exhausted:

Triggered by extreme stamina depletion. Reduces regen for short period.

13. Blueprint System Spec - Survival Stats Component

Blueprint Name: BP_SurvivalStatsComponent Type: Actor Component Attached To: Player Character and NPC Survivors

13.1 Variables

Health

MaxHealth

Stamina

MaxStamina

Hunger

MaxHunger

Thirst

MaxThirst

HungerDrainRate

ThirstDrainRate

StaminaDrainRate

StaminaRegenRate
StarvationDamageRate
DehydrationDamageRate
BoundaryDamageRate
blsSprinting blsExhausted blsBleeding blsPoisoned blsDead

13.2 Functions

DrainHunger(float DeltaTime)
DrainThirst(float DeltaTime)
DrainStamina(float DeltaTime)
RegenerateStamina(float DeltaTime)
ApplyDamage(float DamageAmount, Name DamageType) Heal(float HealAmount) ConsumeFood(float HungerRestoreAmount) DrinkWater(float ThirstRestoreAmount) ApplyBleeding() StopBleeding()
ApplyPoison() CurePoison() CheckDeath()

13.3 Events / Dispatchers

OnHealthChanged
OnStaminaChanged
OnHungerChanged
OnThirstChanged
OnDeath
OnStatusEffectAdded
OnStatusEffectRemoved

13.4 Definition of Done

The Survival Stats system is complete when:

Health, stamina, hunger, and thirst update in real time. Sprinting drains stamina.

Hunger/thirst drain over time. Zero hunger/thirst damages health. Player can die. HUD updates correctly. The same component can be used on NPC Survivors.

14. Inventory System

14.1 Inventory Design

The inventory should be simple but flexible.

Initial design:

Slot-based inventory. Backpack grid. Hotbar slots. Stackable items. Item weight optional later.

Suggested v1 settings:

Backpack: 5x4 grid = 20 slots. Hotbar: 6 slots. Stack size varies by item type.

14.2 Item Categories

Weapons

Tools

Food
Water
Medicine
Crafting Materials
Traps
Ammo / Projectiles
Quest/Lore Items later

14.3 Item Actions

Pick up. Drop. Use. Equip. Move to hotbar. Split stack later.
Inspect later.

14.4 Looting

Loot sources:

The Core crates. Supply drops. Dead Survivors. Abandoned bags. Hidden caches. Animal carcasses.
Plants/resources.

15. Blueprint System Spec - Inventory Component

Blueprint Name: BP_InventoryComponent Type: Actor Component Attached To: Player Character and NPC Survivors

15.1 Structs

ST_ItemData

Fields:

ItemID

ItemName

ItemDescription

ItemType

ItemIcon

MaxStackSize

CurrentStackAmount

blsStackable bCanEquip bCanUse bCanDrop ItemWeight ItemRarity

ST_InventorySlot

Fields:

SlotIndex

ItemData

blsOccupied

15.2 Variables

InventorySlots

HotbarSlots

MaxInventorySlots
MaxHotbarSlots
SelectedHotbarIndex
CurrentEquippedItem

15.3 Functions

AddItem(ST_ItemData ItemToAdd) RemoveItem(int SlotIndex, int Amount) DropItem(int SlotIndex)
UseItem(int SlotIndex) EquipItem(int SlotIndex) MoveItem(int FromSlot, int ToSlot) AssignToHotbar(int
InventorySlot, int HotbarSlot) SelectHotbarSlot(int HotbarIndex) HasItem(Name ItemID, int
RequiredAmount) ConsumeItem(Name ItemID, int Amount)

15.4 Events / Dispatchers

OnInventoryChanged
OnHotbarChanged
OnItemEquipped
OnItemUsed
OnItemDropped

15.5 Definition of Done

Inventory is complete when:

Player can pick up items. Items appear in backpack UI. Items can be assigned to hotbar. Hotbar
selection works with number keys. Items can be used/dropped. Inventory updates after crafting and
looting.

16. Hotbar System

16.1 Purpose

The hotbar gives quick access to important tools and weapons during survival.

Suggested slots:

Weapon/tool
Weapon/tool
Food/water
Medical
Trap/utility

Flexible item

16.2 Hotbar Rules

Hotbar references items in inventory. Empty slot displays blank icon. Selected slot highlights. Using
item removes/updates item stack if consumable. Dropping item removes from hotbar.

17. Crafting System

17.1 Crafting Design

Crafting should be useful but not overcomplicated.

Two types:

Hand Crafting - simple items crafted anywhere. Context Crafting - items requiring fire, water, carcass, or specific resource location. No permanent base building.

17.2 Basic Crafting Materials

Wood

Stone

Vine/Fibre

Bone

Hide

Feathers

Herbs

Resin

1. 2. 3. 4. 5. 6. 1. 2.

Metal scrap

Cloth

17.3 Basic Recipes

Item

Required Materials

Purpose

Wooden Spear

Wood + Stone + Fibre

Basic melee / throwing weapon

Stone Knife

Stone + Fibre

Cutting, harvesting, melee Short Bow Wood + Fibre Ranged weapon Arrows Wood + Stone + Feathers
Bow ammo Bandage Cloth / Fibre + Herbs Stops bleeding Herbal Poultice Herbs + Water Minor healing
Snare Trap Wood + Fibre Small animal trap Noise Maker Scrap + Stone Distraction item Torch Wood +
Cloth + Resin Night visibility Water Pouch Hide + Fibre Carry water Splint Wood + Fibre Leg injury
treatment

17.4 Crafting UI

Crafting screen should show:

Recipe list. Item icon. Required materials. Owned materials. Craft button. Crafting time. Result quantity.

17.5 Crafting Time

Some items craft instantly. Others take time and expose the player.

Examples:

Bandage: quick. Spear: medium. Bow: longer. Campfire: longer.

18. Blueprint System Spec - Crafting Component

Blueprint Name: BP_CraftingComponent Type: Actor Component Attached To: Player Character and possibly NPC Survivors later

18.1 Structs

ST_CraftingRecipe

Fields:

RecipeID

RecipeName

ResultItem

ResultAmount

RequiredItems

CraftingTime

bRequiresFire bRequiresTool RequiredToolID

ST_RequiredItem

Fields:

ItemID

RequiredAmount

18.2 Variables

KnownRecipes

CurrentCraftingRecipe

CraftingProgress

bIsCrafting

18.3 Functions

CanCraft(RecipeID)

StartCrafting(RecipeID)

CancelCrafting()

CompleteCrafting()

ConsumeRecipeItems(RecipeID)

UnlockRecipe(RecipeID)

18.4 Events / Dispatchers

OnCraftingStarted

OnCraftingProgressUpdated

OnCraftingCompleted

OnCraftingCancelled

OnRecipeUnlocked

18.5 Definition of Done

Crafting is complete when:

Player can open crafting UI. Recipe requirements display correctly. Player can craft at least 5 test items. Materials are consumed. Crafted items appear in inventory. Crafting progress can be cancelled.

19. Interaction System

19.1 Interaction Design

The player interacts with world objects using a first-person line trace.

Interactable objects include:

Loot crates. Supply bags. Corpses. Plants. Animal carcasses. Water sources. The Gate. Core containers. Traps. Temporary campfire.

19.2 Interaction Types

Tap interact. Hold interact. Context interact.

Examples:

Tap E: pick up item. Hold E: harvest carcass. Hold E: open heavy crate. Tap E: drink from water source.

20. Blueprint System Spec - Interaction Interface

Blueprint Name: BPI_Interactable Type: Blueprint Interface

20.1 Interface Functions

GetInteractPrompt()

CanInteract(Interactor)

Interact(Interactor)

StartHoldInteract(Interactor)

CancelHoldInteract(Interactor)

CompleteHoldInteract(Interactor)

20.2 Player Interaction Variables

In Player Character:

InteractionDistance

CurrentInteractable

blsHoldingInteract HoldInteractProgress RequiredHoldTime

20.3 Definition of Done

Interaction is complete when:

Player can look at an object and see prompt. Tap interactions work. Hold interactions show progress bar. Line trace is reliable. Interact prompts update correctly.

21. Combat System

21.1 Combat Philosophy

Combat should be tense, risky, and survival-driven. Direct fights should consume stamina, cause injury, and attract attention.

The player should often ask:

"Is this fight worth it?"

21.2 Weapon Types

Initial weapon types:

Fists

Stone knife

Wooden spear

Axe

Bow

Throwing spear

Throwing stone

Future weapon types:

Crossbow

Machete

Makeshift shield

Fire trap

Poison arrows

21.3 Damage Types

Blunt

Slash

Pierce

Fire

Poison

Fall

Boundary

Wildlife

21.4 Melee Combat

Melee actions:

Light attack.

Heavy attack. Block/parry later. Push/shove later.

Stamina cost:

Light attack: low. Heavy attack: medium/high. Blocking: stamina drain when hit.

21.5 Ranged Combat

Ranged actions:

Aim. Draw bow. Release arrow. Throw spear/stone. Ranged should be useful but require skill.

21.6 Stealth Attacks

Stealth attacks can be added later.

Conditions:

Player behind target. Player crouched or undetected. Melee weapon equipped. Target not alerted.

22. Blueprint System Spec - Combat Component

Blueprint Name: BP_CombatComponent Type: Actor Component Attached To: Player Character and NPC Survivors

22.1 Variables

CurrentWeapon

AttackDamage

HeavyAttackDamage

AttackRange

AttackCooldown

StaminaCostLight

StaminaCostHeavy

bCanAttack bIsAttacking bIsBlocking LastDamageDealer

22.2 Functions

LightAttack()

HeavyAttack()

Block()

StopBlock()

ApplyWeaponDamage(TargetActor)

CalculateDamage()

EquipWeapon(ItemData)

UnequipWeapon()

CanPerformAttack()

22.3 Events / Dispatchers

OnAttackStarted

OnAttackHit

OnAttackEnded

OnWeaponEquipped

OnDamageDealt

22.4 Definition of Done

Combat is complete when:

Player can attack with basic melee weapon. NPC can receive damage. NPC can die. Stamina cost applies. Weapon damage values are data-driven. Basic hit detection is reliable.

23. Stealth System

23.1 Stealth Philosophy

Stealth should be practical and survival-focused. It should allow weaker players to avoid stronger Survivors.

Stealth is based on:

Visibility.

Noise. Distance. Movement speed. Light/fire use. Terrain cover. Weather.

23.2 Player Noise

Actions that create noise:

Sprinting. Jumping. Attacking. Breaking branches. Opening crates. Crafting loudly. Animal panic. Campfires/objects.

23.3 Visibility

Player visibility affected by:

Standing/crouching. Bushes/foilage. Time of day. Weather. Light sources. Distance.

23.4 Detection States

NPC detection states:

Unaware

Suspicious

Investigating

Alerted

Engaged

Searching

Fleeing

24. Blueprint System Spec - Stealth Component

Blueprint Name: BP_StealthComponent Type: Actor Component Attached To: Player and NPC Survivors

24.1 Variables

CurrentNoiseLevel

VisibilityLevel

bIsCrouching bIsInFoliage bIsUsingLight NoiseMultiplier VisibilityMultiplier

24.2 Functions

GenerateNoise(float NoiseAmount, float NoiseRadius) CalculateVisibility() SetInFoliage(bool bNewValue) SetUsingLight(bool bNewValue) GetStealthScore()

24.3 Definition of Done

Stealth is complete when:

Sprinting creates detectable noise. Crouching lowers visibility/noise. NPCs investigate noise. Foliage reduces detection. Detection states update clearly.

25. Hunting and Wildlife

25.1 Wildlife Purpose

Wildlife provides food, materials, danger, and environmental life. Animals should not only be resources. Some should be threats.

25.2 Animal Types

Small Prey

Examples:

Rabbits. Birds. Small lizards.

Purpose:

Food. Feathers. Light crafting materials. Medium Prey

Examples:

Deer-like creatures. Boar-like creatures.

Purpose:

Meat. Hide. Bone. Tracking gameplay. Predators

Examples:

Wild dogs. Big cats. Arena Bioforms.

Purpose:

Threat. Area denial. Late-game event pressure.

25.3 Animal Harvesting

Harvestable materials:

Meat.

Hide. Bone. Fat. Feathers.

Harvesting should require:

Knife or sharp tool for best yield. Hold interaction. Time exposure risk.

26. Harvesting System

26.1 Harvestable Resources

Resource nodes:

Trees / branches. Stones. Fibre plants. Herbs. Berries. Mushrooms. Water sources. Animal carcasses.

26.2 Harvest Rules

Some resources require tools. Some can be gathered by hand. Some are dangerous if misidentified. Some respawn per match only, not dynamically.

26.3 Resource Quality

Later system:

Poor quality. Normal quality. High quality. Quality can affect crafting durability and food value.

27. NPC Survivor System

27.1 Design Goal

NPC Survivors are the heart of replayability. They must feel like competitors, not just enemies.

They should:

Loot. Flee. Hide. Hunt. Craft. Track. Ambush. Fight each other. React to events. Change behaviour based on health/resources.

27.2 NPC Count

Standard match:

24 total Survivors. 1 player. 23 NPC Survivors.

Prototype:

1 player. 3-5 NPC Survivors.

Vertical slice:

1 player. 8-12 NPC Survivors.

Full version:

1 player. 23 NPC Survivors.

27.3 NPC Archetypes

The Aggressor

Behaviour:

Rushes The Core. Seeks combat early. Uses melee weapons. High courage, low caution.

Weakness:

Can take early damage. May overcommit. The Hunter

Behaviour:

Tracks animals and Survivors. Uses bow/spear. Moves through forest zones. Good perception.

Weakness:

Avoids close combat. The Runner

Behaviour:

Avoids fights. Loots quickly. High stamina. Survives long by movement.

Weakness:

Poor combat confidence. The Trapper

Behaviour:

Places traps near loot, water, paths, and bodies. Avoids direct fights.

Uses bait.

Weakness:

Vulnerable when exposed. The Stalker

Behaviour:

Follows targets at distance. Attacks when target is weak. Uses stealth.

Weakness:

Avoids groups and open areas. The Scavenger

Behaviour:

Loots dead Survivors. Searches supply zones. Avoids combat unless target is weak.

Weakness:

Can be baited by loot. The Medic

Behaviour:

Prioritises healing items. Avoids conflict. Survives through recovery.

Weakness:

Lower aggression. The Brute

Behaviour:

Strong melee fighter. Low stealth. Intimidates weaker Survivors.

Weakness:

Loud, slower, visible. The Coward

Behaviour:

Hides often. Avoids The Core. Runs from combat. May survive surprisingly long.

Weakness:

Very poor combat. The Strategist

Behaviour:

Waits, observes, chooses safe engagements. Uses terrain and timing. Likely late-game threat.

Weakness:

Slower start.

27.4 NPC Needs

NPC Survivors should have simplified survival stats:

Health. Stamina. Hunger. Thirst. Threat awareness. Inventory. Current goal. They do not need the same complexity as the player at first, but they should behave as if they are surviving.

27.5 NPC Goal States

Possible AI goals:

Rush Core.

Flee Core. Find food. Find water. Find weapon. Hunt animal. Craft item. Hide. Track target. Engage target. Flee target. Loot corpse. Move to safe zone. Investigate noise. Avoid event zone.

28. Blueprint System Spec - NPC Survivor AI

Blueprint Names:

BP_NPCSurvivorCharacter
BP_NPCSurvivorController
BB_NPCSurvivor
BT_NPCSurvivor

28.1 NPC Variables

SurvivorID
SurvivorName
Archetype
AggressionLevel
FearLevel
CautionLevel
HungerPriority
ThirstPriority
CombatConfidence
PreferredWeaponType
CurrentGoal
CurrentTarget
LastKnownTargetLocation
KnownLootLocations
KnownWaterLocations
bHasWeapon bIsInCombat bIsFleeing bIsInjured bIsAlive

28.2 Blackboard Keys

TargetActor
TargetLocation
CurrentGoal
SafeLocation
LootLocation
WaterLocation
ThreatActor
LastNoiseLocation
IsHungry
IsThirsty
IsInjured
HasWeapon
ShouldFlee

28.3 AI Functions

ChooseStartingStrategy()
EvaluateNeeds()
ChooseNextGoal()
FindNearestLoot()
FindNearestWater()
FindNearestCover()

EvaluateCombatTarget(Target)
FleeFromThreat(Threat)
InvestigateNoise(Location)
LootActor(TargetLoot)
UseHealingItem()
UpdateSurvivorState()

28.4 Events / Dispatchers

OnNPCDeath
OnNPCSpottedTarget
OnNPCHeardNoise
OnNPCChangedGoal
OnNPCEnteredCombat
OnNPCFledCombat

28.5 Definition of Done

NPC Survivor AI is complete for prototype when:

At least 3 NPC Survivors spawn. NPCs choose different opening strategies. NPCs can pick up loot.

NPCs can attack player. NPCs can take damage and die. NPCs can fight each other. Survivor count updates when NPC dies.

29. Arena Event System

29.1 Purpose

Arena events prevent the match from becoming static. They create pressure, movement, and unexpected survival choices.

29.2 Event Types

Supply Drop

A supply pod lands in a marked zone.

Effects:

Attracts NPCs. Creates conflict. Offers rare loot. Weather Shift

Examples:

Heavy rain. Fog. Heat surge. Wind storm.

Effects:

Changes visibility. Affects noise. Affects stamina/hunger/thirst. Boundary Surge A region near the outer Dome becomes dangerous.

Effects:

Pushes Survivors inward.

Damages players who stay too long. Drone Scan A drone sweeps an area.

Effects:

Reveals movement. Triggers alarms. Marks Survivors briefly. Predator Release A dangerous animal/Bioform enters a zone.

Effects:

Area denial. Forces movement. Creates chaos. Core Reopening The Core unlocks new containers during mid/late match.

Effects:

Pulls Survivors back to centre. Creates conflict.

29.3 Event Timing

Prototype:

Manual trigger via debug key.

Vertical slice:

Random event timer. Event warning broadcast. Event executes after short delay.

Full version:

Dynamic director chooses events based on Survivor spread, remaining count, and match pace.

30. Blueprint System Spec - Arena Event Manager

Blueprint Name: BP_ArenaEventManager Type: Actor or Game State System

30.1 Variables

ActiveEvent

EventQueue

EventFrequencyMin

EventFrequencyMax

TimeUntilNextEvent

RemainingSurvivors

MatchPhase

ValidEventZones

bEventActive

30.2 Structs

ST_ArenaEventData

Fields:

EventID

EventName

EventType

EventWarningText

EventDuration

TargetZone

bRequiresWarning EventIntensity

30.3 Functions

StartEventTimer()
ChooseRandomEvent()
ChooseTargetZone()
BroadcastEventWarning(EventData)
ActivateEvent(EventData)
EndActiveEvent()
ForceEvent(EventID)
UpdateMatchPhase()

30.4 Events / Dispatchers

OnEventWarning
OnEventStarted
OnEventEnded
OnMatchPhaseChanged

30.5 Definition of Done

Arena Event Manager is complete when:

Events can be triggered manually. Event warnings show on HUD. At least one supply drop works. At least one weather event works. At least one danger zone works. NPCs can react to event zones later.

31. Time of Day and Weather

31.1 Time of Day

A full Trial can use accelerated time.

Possible match length options:

Quick Trial: 20-30 minutes. Standard Trial: 45-90 minutes. Long Trial: 2+ hours later.

Day/night should affect:

Visibility. NPC behaviour. Wildlife danger. Temperature later. Stealth. Fire/light usage.

31.2 Weather

Weather should be controlled by the Dome.

Possible weather states:

Clear. Fog. Rain. Wind. Heat surge.

Electrical storm.

Weather effects:

Rain reduces fire usefulness. Fog reduces sightlines. Wind masks movement noise. Heat increases thirst drain. Storms make the Boundary Field unstable.

32. UI / HUD

32.1 HUD Goals

The HUD should be clean, tense, and survival-focused. It should display enough information without feeling like an arcade shooter.

32.2 HUD Elements

Health bar. Stamina bar. Hunger icon/bar. Thirst icon/bar. Hotbar. Interaction prompt. Remaining Survivors count. Mini-map or compass. Current zone code. Arena event warning. Notification feed.

32.3 Suggested HUD Labels

For the inventory/status section:

Arena Telemetry

Field Status

Survivor Readout

Dome Feed

Trial Data

Recommended:

Arena Telemetry

32.4 Survivor Counter

Example:

Survivors Remaining: 17/24 Remaining: 17 Dome Status: 17 Active

32.5 Event Warnings

Examples:

"Boundary Surge detected in Zone 06." "Drop inbound: Pine Belt." "Drone scan active." "Core access opening in 60 seconds."

33. Mini-map / Compass System

33.1 Mini-map Purpose

The mini-map helps the player understand zone codes and basic navigation without revealing too much. It should not show all NPCs by default.

33.2 Mini-map Features

Player position. Zone code. Core marker. Event zone marker. Supply drop marker. Boundary warning. Last known noise marker later.

33.3 Compass Alternative

A compass may be easier for early prototype.

Compass shows:

Cardinal direction. Core marker. Drop marker. Gate marker in final phase. Zone code text.

Recommendation:

Start with compass + zone code, then add mini-map later.

34. Blueprint System Spec - HUD and Widgets

34.1 Main Widgets

WBP_HUD
WBP_Inventory
WBP_Hotbar
WBP_Crafting
WBP_NotificationFeed
WBP_InteractionPrompt
WBP_MatchStatus
WBP_PauseMenu
WBP_DeathScreen
WBP_VictoryScreen

34.2 HUD Variables

PlayerHealthPercent

PlayerStaminaPercent

PlayerHungerPercent

PlayerThirstPercent

RemainingSurvivors

CurrentZoneCode

CurrentEventText

SelectedHotbarSlot

34.3 Definition of Done

HUD is complete when:

Player vitals update live. Hotbar displays selected slot. Interaction prompt appears/disappears. Survivor count updates. Event warnings show. Death/victory screens trigger correctly.

35. Save / Load System

35.1 Save Philosophy

Because this is a match-based survival game, saving should be carefully controlled.

Options:

No saving during Trial, only settings/progression. Save and quit during Trial, deletes save when loaded. Full save/load for accessibility.

Recommended for early development:

Save settings and meta data first. Add save-and-quit for Trial later.

35.2 Save Data

Save file can store:

Settings. Unlocked difficulty options. Completed Trials. Best survival time. Win/loss stats. Player cosmetics later.

If Trial saving is added:

Player stats. Inventory. NPC states. Dead Survivors. Arena events. Time of day. Loot states.

36. Audio and Atmosphere

36.1 Audio Pillars

Audio must create tension, awareness, and dread. 1. 2. 3.

Important sound layers:

Wind through trees. Distant screams/combat. Wildlife calls. Footsteps on different surfaces. Rustling bushes. Dome hum. Boundary Field crackle. Broadcast announcements. Heartbeat/exhaustion. Rain/storm/fog ambience.

36.2 Music Direction

Music should be minimal and reactive.

Possible states:

Elevator intro tension. Opening rush pulse. Exploration ambience. Combat stingers. Event warning drones. Final phase intensity. Victory silence/resolution.

36.3 Operator Broadcasts

Broadcasts should be cold, minimal, and unsettling.

Examples:

"Trial sequence active." "Twenty-four Survivors registered." "Boundary Field online." "Drop inbound." "Survivor eliminated." "Final phase initiated." "Ascendant confirmed."

37. Visual Direction

37.1 Art Style

Recommended style:

Stylized-realistic survival tone. Not hyper-realistic. Strong silhouettes. Clear environmental readability. Slightly dystopian sci-fi over natural wilderness. This should help performance and production speed.

37.2 Dome Visual Identity

The Dome should mix natural and artificial elements:

Natural forest, marsh, rocks, ruins. Artificial sky shimmer. Distant boundary wall distortion. Metallic Core at centre. Drones/sensors. Hidden speakers/lights. Event towers or beacons.

37.3 The Core Visual Identity

The Core should not be a horn-shaped loot structure. It should be original.

Possible design:

Circular mechanical platform. Central vertical tower. Supply pods arranged in rings. Locking mechanisms. Red/white warning lights. Rising panels. Rotating shutters.

38. Difficulty Settings

38.1 Difficulty Tiers

Easy

Slower hunger/thirst drain. Less aggressive NPCs. More loot. Fewer arena events. More healing items. Normal Balanced survival and combat. Standard NPC aggression. Standard loot. Standard event frequency. Hard Faster hunger/thirst drain. Aggressive NPCs. Scarce loot. More wildlife danger. More frequent events. Brutal / Survivalist Very limited loot. Smarter NPCs. Severe injuries. Dangerous weather. No mid-Trial save.

39. Progression and Replayability

39.1 Meta Progression

Initial version does not need heavy progression.

Possible later unlocks:

New Survivor skins. Starting cosmetic variations. Lore files.

Arena variants. Custom Trial settings. Difficulty badges. Avoid pay-to-win style systems.

39.2 Match Replayability

Replayability comes from:

NPC archetype variation. Random loot distribution. Random drops. Dynamic weather. Arena events. Different starting choices. Hunting/crafting outcomes. Final Survivor combinations.

40. Development Priorities

The project should follow a phased development model. Each phase should have a clear Definition of Done. The first goal is not a full game. The first goal is a playable prototype proving that the core loop is fun.

Phase A - Project Setup and Foundation

A1. Unreal Project Setup

Tasks:

Install Unreal Engine 5.7. Create Blueprint-first project. Add NoGlyph Blueprint Stack to the Unreal project workflow and connect Codex as the primary AI development agent. Configure project settings. Set up Enhanced Input. Set up Git repository. Add .gitignore for Unreal. Create test map.

Definition of Done:

Project launches. Test map opens.

Player can spawn. Git repo is clean. NoGlyph Blueprint Stack is installed, Codex is connected to the Unreal project, and the AI-assisted Unreal workflow is ready.

A2. Folder Structure

Suggested folders:

/Blueprints

/Player

/AI

/Components

/Combat

/Survival
/Inventory
/Crafting
/Interaction
/World
/Arena
/Events
/UI
/SaveLoad
/Audio
/Items
/Data
/Maps
/Test
/Arena
/UI
/Widgets
/Materials
/Audio
/Music
/SFX
/Ambience
/Art
/Characters
/Environment
/Props
/Data
/Items
/Recipes
/NPCs
/Events

Definition of Done:

Folders created. Naming convention agreed. Test content placed correctly.

Phase B - Core Player Systems

B1. Player Character

Tasks:

Create BP_PlayerSurvivorCharacter . Add first-person camera. Add movement. Add sprint. Add crouch. Add jump. Add basic camera effects.

Definition of Done:

Player can move, look, sprint, crouch, and jump in test map.

B2. Survival Stats

Tasks:

Create BP_SurvivalStatsComponent . Add health, stamina, hunger, thirst. Add hunger/thirst drain. Add stamina sprint drain/regen. Add death event. Connect to HUD.

Definition of Done:

All stats update in real time and can cause death.

B3. Minimal HUD

Tasks:

Create WBP_HUD . Display health, stamina, hunger, thirst.

Display remaining Survivors placeholder. Display zone placeholder.

Definition of Done:

HUD updates live during gameplay.

B4. Interaction System

Tasks:

Create BPI_Interactable . Add line trace from camera. Add interact prompt. Add hold progress bar.

Create test pickup and crate.

Definition of Done:

Player can interact with test objects reliably.

B5. Inventory and Hotbar

Tasks:

Create BP_InventoryComponent . Create item data struct. Create inventory UI. Create hotbar UI. Add pick up/drop/use/equip.

Definition of Done:

Player can pick up items, see them in inventory, assign to hotbar, and use/drop them.

Phase C - Core Survival Gameplay

C1. Crafting System

Tasks:

Create BP_CraftingComponent . Add recipe data struct. Add crafting UI. Add at least 5 recipes.

Consume materials and create items.

Definition of Done:

Player can craft basic survival items from gathered resources.

C2. Harvesting

Tasks:

Create harvestable resource actors. Add plants, stones, branches. Add carcass harvesting test. Add tool requirement logic.

Definition of Done:

Player can gather resources and use them for crafting.

C3. Food and Water

Tasks:

Add edible items. Add drinkable items. Add water source actor. Add hunger/thirst restoration.

Definition of Done:

Player can eat/drink and recover hunger/thirst.

C4. Basic Combat

Tasks:

Create BP_CombatComponent . Add fists/basic weapon attack. Add damage system. Add hit detection. Add death reaction.

Definition of Done:

Player can damage and kill a test NPC.

Phase D - Arena Foundation

D1. Elevator Intro

Tasks:

Create elevator pod actor. Add countdown. Add rising movement. Add start match event. Add opening audio.

Definition of Done:

Player starts in elevator and rises into test arena.

D2. The Core

Tasks:

Build placeholder Core structure. Add loot containers. Add opening rush layout. Add risk/reward loot placement.

Definition of Done:

Player can choose to loot The Core or flee at match start.

D3. Boundary Field

Tasks:

Create Dome boundary actor/volume. Add warning zone. Add damage zone. Add visual/audio feedback.

Definition of Done:

Player is warned/damaged when approaching or crossing the Boundary Field.

D4. Survivor Count and Match State

Tasks:

Create BP_TrialGameMode . Track total Survivors. Track deaths. Trigger victory when one remains. Trigger death screen when player dies.

Definition of Done:

Match can start, track deaths, and end properly.

Phase E - NPC Survivor Prototype

E1. NPC Survivor Character

Tasks:

Create BP_NPCSurvivorCharacter . Add survival stats component. Add combat component. Add inventory component placeholder.

Definition of Done:

NPC can move, take damage, and die.

E2. Basic NPC AI

Tasks:

Create AI controller. Create behaviour tree. Add patrol/wander. Add chase player. Add attack player. Add flee behaviour.

Definition of Done:

NPC can detect, chase, attack, flee, and die.

E3. NPC Opening Strategies

Tasks:

Add archetype enum. Add starting behaviour selection. Some NPCs rush Core. Some NPCs flee. Some NPCs hide.

Definition of Done:

NPCs behave differently at match start.

E4. NPC vs NPC Combat

Tasks:

Allow NPCs to detect each other. Allow NPCs to fight each other. Update Survivor count on death.

Definition of Done:

NPCs can eliminate each other without player involvement.

Phase F - Arena Events

F1. Event Manager Stub

Tasks:

Create BP_ArenaEventManager . Add manual debug trigger. Add event warning UI. Add event start/end logic.

Definition of Done:

Events can be triggered and shown on HUD.

F2. Supply Drop Event

Tasks:

Create supply drop actor. Add drop marker. Add loot container. Add landing audio/visual.

Definition of Done:

Supply drop appears in arena and can be looted.

F3. Weather Event

Tasks:

Add fog or rain event. Reduce visibility. Change audio. Add HUD warning.

Definition of Done:

Weather changes gameplay visibility/audio.

F4. Danger Zone Event

Tasks:

Create danger zone volume. Add warning before activation. Add damage over time. Add map/compass marker.

Definition of Done:

Zone becomes dangerous and forces movement.

Phase G - Vertical Slice

G1. Small Complete Arena

Tasks:

Build one small Dome map. Include The Core. Include forest zone. Include ridge/rock zone. Include water source. Include boundary field.

Definition of Done:

Player can complete a full match in a small arena.

G2. Full Prototype Match

Tasks:

Add elevator intro. Add 8-12 NPC Survivors. Add survival stats. Add inventory/hotbar. Add crafting. Add basic combat. Add supply drop event. Add victory/death screen.

Definition of Done:

The game can be played from start to death/victory.

G3. Polish Pass

Tasks:

Add better UI feedback. Add audio cues. Add basic animations. Add hit feedback. Add performance optimisation. Add bug fixing.

Definition of Done:

The vertical slice feels playable and understandable.

Phase H - Full Game Expansion

H1. Full 24 Survivors

Tasks:

Scale NPC system to 23 NPCs. Optimise AI updates. Add archetype variety. Add more names/identities.

Definition of Done:

Full match runs with 24 Survivors total.

H2. Expanded Arena

Tasks:

Add full biome layout. Add landmarks. Add resource distribution. Add event zones. Add hidden caches.

Definition of Done:

Arena supports full match pacing.

H3. Advanced AI

Tasks:

Add NPC looting. Add NPC crafting. Add NPC hunting. Add NPC trap usage. Add NPC memory/last known position.

Definition of Done:

NPCs feel like real Survivors rather than simple enemies.

H4. Full UI/UX

Tasks:

Main menu. Settings. Difficulty selection. Match summary. Save settings. Loading screen with tips/lore.

Definition of Done:

Game has a complete frontend flow.

H5. Early Access Polish

Tasks:

Optimise performance. Add Steam integration later. Add achievements later. Add bug reporting flow. Add balancing pass. Add accessibility options.

Definition of Done:

Game is ready for limited external testing.

41. Data Tables

The game should use data tables wherever possible so values are easy to balance.

Recommended Data Tables:

DT_Items
DT_CraftingRecipes
DT_NPCSurvivors
DT_NPCArchetypes
DT_ArenaEvents
DT_LootTables
DT_Wildlife
DT_DifficultySettings

42. Naming Conventions

42.1 Blueprint Prefixes

Prefix

Meaning

BP_

Blueprint Actor/Object

BPI_

Blueprint Interface

WBP_

Widget Blueprint

ABP_

Animation Blueprint

DT_

Data Table

ST_

Struct

E_

Enum

M_

Material

MI_

Material Instance

SFX_

Sound Effect

MUS_

Music

AMB_

Ambient Sound

42.2 Example Names

BP_PlayerSurvivorCharacter

BP_NPCSurvivorCharacter

BP_InventoryComponent

BP_SurvivalStatsComponent

BP_CraftingComponent

BP_CombatComponent

BP_ArenaEventManager

BP_BoundaryFieldVolume

BP_CoreSupplyCrate

WBP_HUD
WBP_Inventory
WBP_CraftingMenu
WBP_DeathScreen
WBP_VictoryScreen

43. Codex + NoGlyph Blueprint Stack Prompting Strategy

The project will use Codex + NoGlyph Blueprint Stack as the main AI-assisted Unreal Engine workflow.

NoGlyph Blueprint Stack acts as the bridge between Codex and Unreal Engine. Codex provides the reasoning, planning, coding, and implementation guidance, while NoGlyph exposes Unreal Editor tools so Codex can inspect assets, create or edit Blueprints, modify variables, work with node graphs, update data tables, create materials, assist with widgets, and automate editor-side tasks.

43.1 Core Workflow

The AI workflow should follow this order:

Jonathan defines the system goal using the GDD. Codex receives the system-specific prompt. NoGlyph exposes the relevant Unreal Editor tools. Codex inspects the existing project assets. Codex creates or edits only the requested Blueprints/assets. Jonathan tests the result in the relevant test map. Bugs are fixed before moving to the next system. The working change is committed to GitHub.

The AI should not be asked to build the entire game at once. Each system must be built, tested, and committed separately.

43.2 Agent Rules

When using Codex through NoGlyph, every prompt should follow these rules:

Work only inside /Game/The24th/ unless explicitly instructed otherwise. Do not touch Engine content. Do not rename or delete existing assets unless explicitly requested. Build one system at a time. Prefer modular Actor Components for reusable gameplay logic. Use existing naming conventions from the GDD. Use existing structs, enums, data tables, and widget names where available. Summarize every change after editing. Tell Jonathan exactly what to test after each change. Do not move to the next system until the current one is tested.

43.3 Prompt Requirements

Each Codex + NoGlyph prompt should include:

The exact Blueprint, widget, data table, map, material, or asset name. The folder path where it should work. The purpose of the system. Required variables. Required functions. Required events or dispatchers. Required UI/widget connections. Which systems it may connect to. Which systems it must not touch. Definition of Done.

43.4 Example Context Prompt

You are connected to my Unreal Engine project through NoGlyph Blueprint Stack.

Project: The 24th Engine: Unreal Engine 5.7 Workflow: Blueprint-first Primary AI Agent: Codex Unreal Bridge: NoGlyph Blueprint Stack

The 24th is a first-person arena survival game where 24 Survivors rise into a sealed Dome and fight to become the last one alive.

Core terms:

- Survivors = contestants
- The Dome = arena
- The Core = central loot zone
- The Trial = match
- The Operators = arena controllers
- Boundary Field = arena border
- Ascendant = winner

Rules:

- Work only inside /Game/The24th/
- Do not touch Engine content.
- Do not delete or rename assets unless I explicitly ask.
- Build one Blueprint system at a time.
- Use modular Actor Components where possible.
- After each change, summarize what changed and what I need to test.

43.5 Example System Prompt

Use NoGlyph Blueprint Stack tools to inspect and edit BP_SurvivalStatsComponent in /Game/The24th/Blueprints/Components/.

Purpose:

This component manages Health, Stamina, Hunger, and Thirst for both the player and NPC Survivors.

Create or verify these variables:

- Health, MaxHealth
- Stamina, MaxStamina
- Hunger, MaxHunger
- Thirst, MaxThirst
- HungerDrainRate
- ThirstDrainRate
- StaminaDrainRate
- StaminaRegenRate
- StarvationDamageRate
- DehydrationDamageRate
- bIsSprinting
- bIsDead

Create or verify these functions:

- DrainHunger
- DrainThirst
- DrainStamina
- RegenerateStamina
- ApplyDamage
- Heal
- ConsumeFood
- DrinkWater
- CheckDeath

Create or verify these event dispatchers:

- OnHealthChanged
- OnStaminaChanged
- OnHungerChanged
- OnThirstChanged
- OnDeath

Requirements:

- Clamp all stat values between 0 and their max value.
- Hunger and thirst should drain over time.
- Sprinting should drain stamina.
- Stamina should regenerate when not sprinting.
- Zero hunger or thirst should damage health over time.
- Health reaching zero should trigger OnDeath.
- Do not create HUD, inventory, crafting, combat, or AI logic in this step.

Definition of Done:

The component compiles, exposes the required variables/functions/dispatchers, and can be tested on the player character in Map_TestSystems.

43.6 Testing and Git Rule

After Codex/NoGlyph edits any system:

Compile the edited Blueprint. Save all modified assets. Test the system in the correct test map. Fix any compile or runtime errors. Commit the working change to GitHub.

Recommended commit style:

- Add survival stats component foundation
- Add inventory component base logic
- Add interaction line trace system
- Add prototype elevator intro sequence

44. First Prototype Scope

The first playable prototype should not attempt to include the full game.

44.1 Prototype Goal

Prove this loop:

Rise into arena -> choose loot or flee -> gather resources -> craft weapon -> fight/avoid NPCs -> survive until victory/death.

44.2 Prototype Features

Must-have:

First-person player. Elevator intro. Small test arena. The Core placeholder.

Health/stamina/hunger/thirst. Inventory/hotbar. 5 items. 3 craftable recipes. Basic melee weapon. 3 NPC Survivors. Survivor count. Death screen. Basic victory condition.

Should-have:

Basic bow. Basic wildlife. Supply drop event. Boundary Field.

Crafting UI.

Nice-to-have:

Mini-map. Weather event. NPC looting. NPC vs NPC combat.

44.3 Prototype Definition of Done

The prototype is complete when:

The player starts in an elevator and rises into the arena. The player can collect items. The player can manage survival stats. The player can craft at least one weapon and one healing item. NPC Survivors can attack and die. The player can die. The player can win when all NPCs are eliminated. The match can restart.

45. Current Open Design Questions

These decisions can be refined later:

Should the game be first-person only, or include third-person toggle later? Should the Dome have a visible crowd/broadcast system or remain mysterious? Should NPC Survivors have names and short backstories? Should the player start empty-handed or with one random item? Should the match include thirst from the start, or only hunger/stamina for prototype? Should The Core remain accessible all match or lock/reopen in phases? Should the Boundary Field only block escape, or actively push players inward? Should the victory require reaching The Gate, or instantly end when one Survivor remains? Should the game include permanent progression, or stay match-based? Should wildlife be normal animals, artificial Bioforms, or a mix?

46. Version Notes

v0.1

Created initial GDD for The 24th. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10.

Included:

Project overview. IP-safe terminology. Core pillars. Gameplay loop. Match structure. Arena design. Survival systems. Inventory/hotbar. Crafting. Combat. Stealth. Hunting/harvesting. NPC Survivor AI. Arena event system. UI/HUD. Audio/visual direction. Development phases. Blueprint-facing system specs. Codex + NoGlyph Blueprint Stack prompting strategy.

47. Overall Definition of Done

The first full version of The 24th is complete when:

Player can start a Trial from the main menu. Player rises into the Dome through the elevator sequence. 24 Survivors exist in the match. 23 NPC Survivors behave according to archetypes. Player can loot, craft, hunt, harvest, eat, drink, fight, hide, and survive. The Dome contains meaningful zones, resources, wildlife, and hazards. Arena events create pressure and replayability. Survivor count updates dynamically. Death and victory states work correctly. The final Survivor can reach The Gate and become the Ascendant. The game loop feels tense, replayable, and original.